

Sampling

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1 In class assignment 9//11/2026

You will submit:

- 1) Answers to the questions
- 2) A screenshot of the R code

Raise your hand if you have any questions.

2 PArt 1:

Estimate the Standard Error by hand (you can use a calculator) from the systolic blood dataset. You have the standard deviation and the n , so it should be pretty straightforward!

The equation for the standard error is:

$$SE = \frac{s}{\sqrt{n}}$$

where s is the standard deviation and n is the sample size. # Part 2

2.1 Random sampling

You are working in a small pond in New England. This pond has exactly 100 American eels (*Anguila rostrata*). The mean total length of this population is 100-cm. We will be randomly sampling a small number ($n=12$) and estimating the mean of the population. We will also be estimating the standard error. In this case, we know the real population mean (100-cm) and thus, we can tell whether our mean estimate was close or not to the real population mean!



Figure 2.1: American Eel

We will be doing this pretty straightforward exercise in program R.

On Canvas there should be an `eel.csv` file for program R.

2.1.1 Program R

You can import the dataset using the function `read.csv()`

Type this into R:

```
eel<-read.csv('eel.csv')
```

The text within the brackets, and contained in quotation marks, is the file path (directory and file name) of the file you want to import. This is my directory. Yours would most likely be different. Perhaps something like...

If you want to look at the top five rows of your head you can use the following:

```
head(eel)
```

	Individual	Length
1	1	109.63328
2	2	98.78224
3	3	112.82370
4	4	116.94909
5	5	117.45646
6	6	93.83629

We will do the same exact analysis we did in excel. First, we need to confirm the population mean:

```
mean(eel$Length)
```

```
[1] 100
```

Then, let's sample 12 individuals using:

```
sampld.eels<-sample(eel$Length,12)
```

You need to estimate the mean using `mean`, the standard deviation using `sd`, and the confidence interval using the following equation:

$$CI = mean \pm 2 \times std.err$$

So, to estimate the mean use:

```
mean<-mean(sampled.eels)
std.dev<-sd(sampled.eels)
```

And to estimate the standard error you use:

$$std.err = st.dev \div \sqrt{n}$$

And remember that n is 12.

Answer:

Did the confidence interval include the real mean of 100?

Now, do the same exercise 10 times. You can rerun the code, without rewriting it. You should get a different result each time! Submit a screenshot of the results and answer: how many times did the CI include the actual mean of 100?